



COMPUTER AWARENESS

- Which one of the following boolean algebraic rule is correct?
(a) $A \cdot A' = 1$ (b) $A + AB = A + B$
(c) $A + A'B = A + B$ (d) $A(A + B) = B$
- The representation of a floating point binary number +1001.11 in 8-bit fraction and 6-bit exponent format is
(a) fraction : 01001110 exponent: 000100
(b) fraction : 00001001 exponent: 000011
(c) fraction : 10010000 exponent: 110000
(d) fraction : 00100100 exponent: 011000
- Which term is redundant in the expression $AB + A'C + BC$?
(a) $B'C$ (b) $A'C$ (c) AB (d) None
- Let the memory access time is 10 milliseconds and cache access time is 10 microsecond. Assume the cache hit ratio 15%. The effective memory access time is
(a) 2 milliseconds (b) 18.51 milliseconds
(c) 1.85 microsecond (d) 1.85 milliseconds
- Which of the following is the representation of decimal number (-147) in 2's complement notation on a 12 bit machine?
(a) 111101101100 (b) 110001001101
(c) 111101101101 (d) 000001101101
- The first instruction of bootstrap loader program of an operating system is stored in
(a) RAM (b) Hard disk
(c) BIOS (d) None of these
- Consider the equation $(123)_5 = (x8)_y$ with x and y as unknown. The number of possible solutions is
(a) 1 (b) 2 (c) 3 (d) 4
- The smallest integer that can be represented by an 8 bit number in 2's complement form is
(a) -256 (b) -128 (c) -127 (d) -255
- Which of the following in a functionally complete set of gates?
I. NAND
II. NOR
(a) I but not II (b) II but not I
(c) Neither I nor II (d) Both I and II
- The total number of binary function that can be defined using n boolean variables is
(a) 2^{n-1} (b) 2^n (c) 2^{n+1} (d) None

ANSWER

1.	C	2.	A	3.	D	4.	B	5.	C
6.	A	7.	C	8.	B	9.	D	10.	D

SOLUTION

- (c) $A \cdot A' = 0$
So, option (a) is wrong
 $A + AB$
 $A(1 + B) = A$

So, option (b) is wrong

$$A + A'B$$

$$(A + A')(A + B)$$

$$1(A + B) = A + B$$

So, option (c) is right.

$$A(A + B)$$

$$A \cdot A + B = A + B$$

So, option (d) is wrong.

2. (a)

The binary number + 1001.11 is represented with an 8-bit fraction and 6-bit exponent as follows

Fraction	Exponent
01001110	000100

The fraction has a 0 in the left most position to denote positive. The binary point of the fraction follows the sign bit but is not shown in the register. The exponent has the equivalent binary number +4. The floating point number is equivalent to $m \times 2^e = +(.1001110)_2 \times 2^{+4}$

3. (d)

$$AB + A'C + BC = AB + A'C + BC(A + A')$$

[by Complementary law]

$$= AB + A'C + ABC + A'BC$$

$$= AB + ABC + A'C + A'BC$$

[By Associative law]

$$= AB(1 + C) + A'C(1 + B) \text{ [by Absorption law]}$$

$$= AB + A'C$$

4. (*) Hit Ratio = 15%

Cache memory time = 10 microseconds

Memory access time = 10 milliseconds

Effective Memory Access Time (EAT) = Hit Ratio

(Cache memory time + Memory access time) + (sign hit)*

(Cache memory time + 2* Memory access time)

$$EAT = 0.15(0.01) + 10 + (1 - 0.15)(0.01 + 2 \times 10)$$

$$= (0.15)(10.01) + (0.85)(20.01)$$

$$= 1.5015 + 17.0085 = 18.51 \text{ milliseconds}$$

None option is correct.

5. (c)

$(147)_{10}$ convert into $(?)_2$

2	147	
2	73	1
2	36	1
2	18	0
2	9	0
2	4	1
2	2	0
	1	0

$$(147)_{10} = (10010011)_2$$

First bit (the left most) indicates the sign

1 = negative, 0 = Positive

Positive binary computer representation on 12 bit if needed, add extra 0, in front of the base 2 member:

$$\text{So, } (147)_{10} = 000010010011$$

For 1's complement, replace all the bits on 0 with 1s and all the bits set on 1 with 0s.

1's complement of

$$(-147) = 111101101100$$





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For 2's complement, add 1 to the number of 1's complement

2's complement of (-147)

111101101100

+ 1

111101101101

6. (a) RAM.

7. (c)

Changing (123) base 5 into base 10.

$$= 1 \times 5^2 + 2 \times 5^1 + 3 \times 5^0$$

$$= 1 \times 25 + 2 \times 5 + 3 \times 1^0$$

$$= 25 + 10 + 3 = 38 \quad \dots\dots\dots(i)$$

Changing x8 base in decimal

$$= x \times y^1 + 8 \times y^0$$

$$= xy + 8 \quad \dots\dots\dots(ii)$$

From Eqs. (i) and (ii), we get

$$xy + 8 = 38$$

$$\Rightarrow xy = 30$$

So, possible combinations = (1, 30), (2, 15), (3, 10)

But we have 8 present in x8, so base y > 8

Therefore, total solution = 3

8. (b)

The range of numbers that can be represented by n-bits in 2's complement form is $(-2)^{n-1}$ to $(2^{n-1}) - 1$.

Hence, the smallest number is $(-2)^{n-1}$

$$= (-2)^{8-1} = (-2)^7 = -128$$

9. (d)

A functionally complete set of Boolean function consists of a set of Boolean functions from which you can construct all Boolean function. {AND, NOT}, {OR, NOT}, {NAND}, {NOR} are four functionally complete sets.

10. (d)

For n-Boolean variables, there are 2^n possible Boolean inputs. Each input can generate either True or False as the output. Hence, total number of different Boolean function = 2^{2^n} .

